

On Greedy Edge Set Partitions of Graphs by Cliques

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Abstract

Clique partitions and clique coverings are widely used in many applications of graph models as classifying and clustering tools, and the related theory is extensively studied in literatures.

In 1966 P. Erdős, A. Goodman, and L. Posa proved that (The representation of a graph by set intersections, Canad. J. Math., 18, pp.106-112) the edge set of any simple graph G with n vertices, no one of which is isolated vertex, can be partitioned using at most $\lfloor n^2/4 \rfloor$ cliques. They also pointed out the correspondence between the edge clique covering and the set representation of multifamily of sets for a graph.

In 1994 McGuinness showed that (Journal of Graph Theory, Volume 18, Issue 4 (1994) pp. 427 - 430) any greedy clique decomposition of a graph of order n has at most $\lfloor n^2/4 \rfloor$ cliques, where a greedy clique decomposition of a graph is obtained by removing maximal cliques from a graph one by one until the graph is empty.

In this paper we studied the set representations for graphs in the variant case of family of distinct sets, including the greedy way to get the corresponding partition of the edge set of the graph. As an immediate corollary, one may obtain the optimal upper bound for the associated intersection numbers of graphs.

1 Background and Introduction

By an *multigraph* $M = (V(M), E(M), q)$, we mean a triple system consisting of a set $V(M)$ of *vertices*, a set $E(M)$ of *edges*, and an integer-valued function q defined on $V(M) \times V(M)$ in the following way. For each unordered pair $\{u, v\} \subset$

$V(M)$, let $q(u, v)$ be the number of *parallel edges* joining u with v . If $q(u, v) \neq 0$, then we say that $\{u, v\}$ is an *edge* of M and $q(u, v)$ is called the *multiplicity* of the edge $\{u, v\}$. For the main results in this paper, we consider only *finite, undirected, simple* multigraphs, where *simple* means that $q(u, v) \leq 1$ for every $\{u, v\} \subset V$ and $q(u, u) = 0$ for every $u \in V(M)$. Therefore we simply call such multigraphs to be *graphs* for short throughout this article, unless otherwise stated.

For a vertex subset $S \subseteq V(M)$, $\langle S \rangle_V$ denotes the *subgraph induced by S* . For a vertex v in M , $d_M(v)$ or $d(v)$ denote the *degree* of v in M . Let $\mathcal{F} = \{S_1, \dots, S_p\}$ be a *family* of distinct nonempty subsets of a set X . Then $\mathbf{S}(\mathcal{F})$ denotes the union of sets in $\mathcal{F}$. The *intersection multigraph* of $\mathcal{F}$, denoted $\Omega(\mathcal{F})$, is defined by $V(\Omega(\mathcal{F})) = \mathcal{F}$, with $|S_i \cap S_j| = q(S_i, S_j)$ whenever $i \neq j$. Of course, so long as we are involved in this paper, $|S_i \cap S_j|$ always equal either 0 or 1 for all $i \neq j$, as appointed above.

We say that a multigraph M is an *intersection multigraph* on a family (a multifamily, respectively) $\mathcal{F}$, if there exist a family (a multifamily, respectively) $\mathcal{F}$ such that $M \cong \Omega(\mathcal{F})$. We say that $\mathcal{F}$ is a *representation* (a *multifamily representation* respectively) of the multigraph M . The *intersection number*, denoted $\omega(M)$ (*multifamily intersection number*, denoted $\omega_m(M)$, respectively), of a given multigraph M is the minimum cardinality of a set X such that M is an intersection multigraph (*multifamily intersection multigraph*, respectively) on a family (a multifamily, respectively) $\mathcal{F}$ consisting of distinct (not necessarily distinct, respectively) subsets of X . In this case we also say that $\mathcal{F}$ is a *minimum representation* (*multifamily representation*, respectively) of M .

Note that given a representation $\{S_v \mid v \in V(M)\}$ of M and a vertex subset $S \subseteq V(M)$, then $\{S_v \mid v \in S\}$ form a representation of $\langle S \rangle_V$. Thus we know that $\omega(M)$ is not less than $\omega(\langle S \rangle_V)$ for any $S \subseteq V(M)$. Similarly for $\omega_m(M)$.

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In 1966 Erdős et al. [1] proved that the edge set of any simple graph G with n vertices, no one of which is isolated vertex, can be partitioned using at most $\lfloor n^2/4 \rfloor$ cliques. A couple of tens of years behind McGuinness [3] proved that any greedy clique partition is such a partition.

A multifamily representation of a graph G is a family of sets each member of which represent a vertex in G and the intersection relation of two members of which represent the adjacency of the two corresponding vertices in G . Erdős et al. [1] suggested a one-one correspondence between multifamily representations and clique coverings of a graph G . In fact, we may define a *multifamily representation of a multigraph M* to be a family of sets for which each member represents a vertex in M , and the two vertices are adjacent with q edges in M if and only if the corresponding representation sets have an intersection of cardinality q . Then there is also a one-one correspondence between multifamily representations and clique partitions of M .

In next section we will narrate this correspondence in full detail. If a multifamily representation of a multigraph M has pairwise distinct member sets, then it is called a *representation* of M . And then we turn the correspondence to prove that any n -vertex graph can be represented by at most $\lfloor n^2/4 \rfloor$ elements and we can accomplish such a representation from any greedy clique partition by a straightforward method based on this correspondence. In the end, certain future directions will be mentioned.

2 Clique Partitions and Set Representations of Graphs

Given a multigraph $M = (V(M), E(M), q)$, $Q \subseteq V(M)$ is said to be a *clique* of M if every pair of distinct vertices u, v in Q has $q(u, v) \neq 0$. A *clique partition* $\mathcal{Q}$ of a multigraph is a set of cliques such that every pair of distinct vertices u, v in $V(M)$ simultaneously appear in exactly $q(u, v)$ cliques in $\mathcal{Q}$ and for each *isolated vertex*, that is, vertex with no edge incident to it, we need to use at least one *trivial clique*, that is, clique with only one vertex, in $\mathcal{Q}$ to cover it. The minimum cardinality of a clique partition of M is called the *clique partition number* of M , and is denoted by $cp(M)$. This number must exist as the edge set of M forms a clique partition for M . We refer to a clique partition of M with the cardinality $cp(M)$ as a *minimum clique partition* of M .

Note that a clique partition $\mathcal{Q}$ of M give rise to a clique partition of $M - v$ by deleting the vertex v from each clique in $\mathcal{Q}$. Thus $cp(M)$ is not less than the clique partition number of any induced subgraph of M .

Now we introduce a one-one correspondence between multifamily representations and clique partitions of a multigraph M as following.

Given a multigraph $M^{(n)} = (V(M), E(M), q)$ with n vertices, we first construct a clique partition

$$\mathcal{Q} = \{Q_1, \dots, Q_p\}$$

of it. Then with each clique Q_k we associate an element e_k and with each vertex v_α we associate a set $S_{\mathcal{Q}}(v_\alpha)$ of elements e_k , where

$$e_k \in S_{\mathcal{Q}}(v_\alpha) \Leftrightarrow v_\alpha \in Q_k,$$

i.e., $S_{\mathcal{Q}}(v_\alpha)$ is the collection of those elements for which the corresponding cliques contains v_α . Thus we obtain

$$\mathcal{F}(\mathcal{Q}) \equiv \{S_{\mathcal{Q}}(v) : v \in V(M)\}.$$

Then clearly

$$\mathbf{S}(\mathcal{F}(\mathcal{Q})) \equiv \bigcup_{v \in V(M)} S_{\mathcal{Q}}(v)$$

contains p elements. And

$$|S_{\mathcal{Q}}(v_\alpha) \cap S_{\mathcal{Q}}(v_\beta)| = q(v_\alpha, v_\beta),$$

since there is exactly $q(v_\alpha, v_\beta)$ cliques simultaneously containing the two vertices v_α, v_β . Thus we have constructed a multifamily representation

$$\mathcal{F}(\mathcal{Q}) = \{S_{\mathcal{Q}}(v) : v \in V(M)\}$$

from the clique partition $\mathcal{Q}$ of M , where

$$|\mathbf{S}(\mathcal{F}(\mathcal{Q}))| \equiv \left| \bigcup_{v \in V(M)} S_{\mathcal{Q}}(v) \right| = p = |\mathcal{Q}|.$$

Conversely, given a multifamily representation $\mathcal{F} = \{S_1, \dots, S_n\}$ of M with vertex set $V(M) = \{v_1, \dots, v_n\}$, where S_α correspond to the set attaching to v_α , then we can also construct a clique partition of M by the following way.

Let

$$\mathbf{S}(\mathcal{F}) \equiv \bigcup_{\alpha=1}^n S_{\alpha} = \{e_1, \dots, e_p\}.$$

For each fixed e_k in $\mathbf{S}(\mathcal{F})$ we form a clique $Q_{\mathcal{F}}(e_k)$ using those vertices v_{α} such that the set S_{α} attaching to it contains e_k . Clearly each $Q_{\mathcal{F}}(e_k)$ is indeed a clique of M . Thus we obtain

$$\mathcal{Q}(\mathcal{F}) = \{Q_{\mathcal{F}}(e_1), \dots, Q_{\mathcal{F}}(e_p)\}.$$

And

$$q(v_{\alpha}, v_{\beta}) = |S_{\alpha} \cap S_{\beta}|$$

which is the number of cliques in $\mathcal{Q}(\mathcal{F})$ simultaneously containing v_{α} and v_{β} . Since each element in S_{α} exactly represent a clique in $\mathcal{Q}(\mathcal{F})$ containing v_{α} . Thus we have constructed a clique partition $\mathcal{Q}(\mathcal{F})$ of M from the multifamily representation $\mathcal{F}$ of M , where

$$|\mathcal{Q}(\mathcal{F})| = p = \left| \bigcup_{\alpha=1}^n S_{\alpha} \right| \equiv |\mathbf{S}(\mathcal{F})|.$$

Thus we have established a one-one correspondence between multifamily representations and clique partitions of the multigraph M .

From above we know that $\omega_m(M) = cp(M)$ and in particular we have the following:

THEOREM 2.1. *Let G be a finite undirected simple graph, we have $\omega_m(G) = cp(G)$.*

In 1994, McGuinness [3] proved the following theorem:

THEOREM 2.2. *Every greedy clique partition of an n -vertex graph uses at most $\lfloor n^2/4 \rfloor$ cliques.*

In above theorem, the greedy clique partition of an n -vertex graph $G^{(n)}$ means an ordered set $\mathbf{Q} = \{Q_1, \dots, Q_m\}$ such that each Q_i is a maximal clique in $G - \bigcup_{j < i} E(Q_j)$, where $G - \bigcup_{j < i} E(Q_j)$ means the subgraph of G obtained by deleting all edges in the edge subset $\bigcup_{j < i} E(Q_j)$ while leaving all vertices in G preserved.

3 Main Results

Here we study the concept on the case of family representations of distinct sets. For a representation $\mathcal{F}$ of G , we call those elements in $\mathbf{S}(\mathcal{F})$ which appear in only one member of $\mathcal{F}$ as *monopolized elements*. The main result is as follows:

THEOREM 3.1. *Every representation $\mathcal{F}$ of a graph $G^{(n)}$ with $n \geq 4$ obtained from $\mathcal{F}(\mathbf{Q})$, where $\mathbf{Q}$ is any greedy clique partition of $G^{(n)}$, by successively attaching monopolized elements to the sets which repetitiously occur in $\mathcal{F}(\mathbf{Q})$. Note that by this method, at most $\lfloor n^2/4 \rfloor$ elements are used.*

Before proving the main theorem, we need the following lemma:

LEMMA 3.2. *For any clique partition $\mathcal{Q}$ of a graph G , $\mathcal{F}(\mathcal{Q}) = \{S_{\mathcal{Q}}(v) : v \in V(G)\}$ has two identical sets, say $S_{\mathcal{Q}}(u), S_{\mathcal{Q}}(v)$, in it only if the clique in $\mathcal{Q}$ simultaneously containing u, v is a maximal clique, say Q_{uv} , in G and has u, v as its monopolized elements. That is, u, v are in no clique of $\mathcal{Q}$ except Q_{uv} , implying that all vertices adjacent to u, v in G are just all vertices in $Q_{uv} - \{u, v\}$.*

Proof. If there is a clique Q' properly containing Q_{uv} in G , say vertex w being in Q' but not in Q_{uv} , then no clique in $\mathcal{Q}$ can simultaneously contain the three vertices u, v, w . Thus the clique in $\mathcal{Q}$ simultaneously containing u, w doesn't contain v and the clique in $\mathcal{Q}$ simultaneously containing v, w doesn't contain u , and therefore we must have $S_{\mathcal{Q}}(u) \neq S_{\mathcal{Q}}(v)$.

If u , say, belong to one clique Q'' in $\mathcal{Q}$ other than Q_{uv} , then there is a vertex, say u' , adjacent to u and not in Q_{uv} . In case that u' is not adjacent to v we must have $S_{\mathcal{Q}}(u) \neq S_{\mathcal{Q}}(v)$. In case that u' is adjacent to v , then no clique in $\mathcal{Q}$ can simultaneously contain u, v, u' . Thus the clique in $\mathcal{Q}$ simultaneously containing u, u' doesn't contain v and the clique in $\mathcal{Q}$ simultaneously containing u', v doesn't contain u , and therefore we must have $S_{\mathcal{Q}}(u) \neq S_{\mathcal{Q}}(v)$. $\square$

Then we are in a position to proceed the proof of theorem 3.1:

Proof. We use induction on n .

When $n = 4$, it is easy to draw all the 11 different graphs on four vertices and to check that every representation of each one of them derived from the statement of theorem 3.1 uses at most $\lfloor n^2/4 \rfloor$ elements. As for $n = 5$, the number of non-isomorphic graphs is sufficiently large to make a reduction being desired. Note that $\lfloor 5^2/4 \rfloor - \lfloor 4^2/4 \rfloor = 6 - 4 = 2$ and therefore we have two new elements in proceeding from $n = 4$ to $n = 5$. If $G^{(5)}$ has one vertex with degree 2 or less, then we reduce $G^{(5)}$ to $G^{(4)}$ by deleting this vertex and all edges incident to it and then obtain one of the 11 graphs with 4 vertices. This vertex form a maximal clique in $G^{(5)}$ with some edge in

$G^{(4)}$ only if $G^{(5)}$ is one of the 13 non-isomorphic graphs in figure 1, where hollow circle denote this vertex and dashed lines denote the edges incident to it.

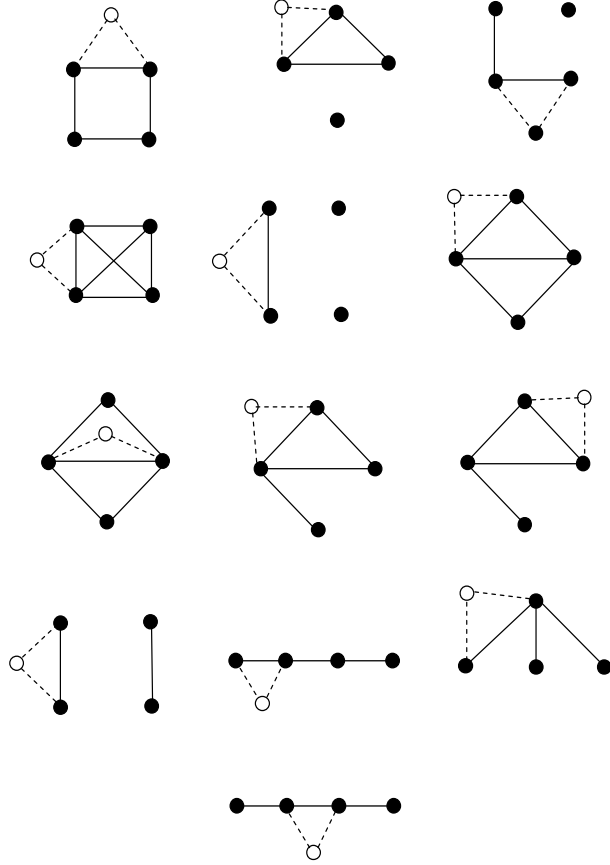


Figure 1:

It is easy to check that every representation of any one of them derived from the statement of theorem 3.1 uses at most $\lfloor 5^2/4 \rfloor = 6$ elements.

As for the case that there is no maximal clique in $G^{(5)}$ simultaneously containing this vertex and some edge in $G^{(4)}$, then in any greedy clique partition of $G^{(5)}$ we must use all the edges incident to this vertex as members of this greedy clique partition. Thus in this case in order to obtain any representation derived from the statement of theorem 3.1, we can first form a representation of $G^{(4)}$ by the statement of theorem 3.1 and then go back to $G^{(5)}$ using the available two new elements to represent at most two edges incident to this vertex and thus we can affirm that in this case all representations of $G^{(5)}$ derived from the statement of theorem 3.1 uses at most $\lfloor 5^2/4 \rfloor = 6$ elements.

As for the case that there is no edge in $G^{(5)}$ incident to this vertex, we can first form a repre-

sentation of $G^{(4)}$ by the statement of theorem 3.1 and then go back to $G^{(5)}$ using one new monopolized element on this vertex.

Due to above, now we need to consider only those multigraphs on five vertices for which every vertex has degree greater than or equal to 3. There are only three such graphs and these are easy to discuss, see figure 2.

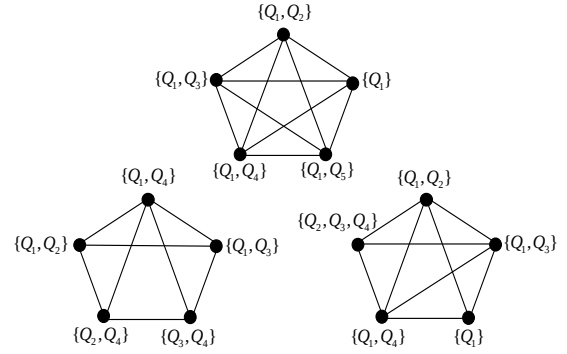


Figure 2:

Thus we have proved the theorem for $n = 4, 5$.

Now let $\mathcal{F}$ be a representation of $G^{(n)}$ with $n \geq 6$ derived from $\mathcal{F}(\mathbf{Q})$, where $\mathbf{Q} = \{Q_1, \dots, Q_m\}$ is a greedy clique partition of $G^{(n)}$, by the statement of theorem 3.1. Note that deleting Q_j from the set $\mathbf{Q}$ leaves a greedy clique partition of $G - E(Q_j)$.

In case that each Q_j has at least three edges, we have $m \leq \binom{n}{2}/3 < n^2/6$. Assume for the time being that every Q_i has exactly three edges, that is, is exactly a triangle. Now if every triangle in $\mathbf{Q}$ has at most one of its three vertices of degree 2, then by lemma 3.2 we needn't use any monopolized element in the statement of theorem 3.1 for this greedy clique partition. If there is a triangle in $\mathbf{Q}$ with at least two of its three vertices of degree 2, then recalling that $G^{(n)}$ have at least six vertices, two vertices of degree 2 in this triangle make m be less than or equal to $(\binom{n}{2}/3) - 2 < (n^2/6) - 2$. Thus despite that we maybe need two more monopolized elements for this triangle, yet in the same time we also have two less cliques (as K_3) in $\mathbf{Q}$. Besides, if there is a clique of cardinality $3 + r$ where $r > 0$ in $\mathbf{Q}$, then despite that we maybe need r more monopolized elements for this clique, yet in the same time by the fact that $\binom{3+r}{2} \geq 3(r+1)$, we also have r less cliques (as K_3) in $\mathbf{Q}$, where note that $\binom{3+r}{2}$ is the number of edges in a clique of cardinality $3 + r$ and $3(r+1)$ is the total number of edges in $r+1$ triangles. In fact, we may need rather $r+1$ or $r+2$ than r more monopolized elements for this clique of cardinality $3 + r$.

By lemma 3.2, we need use $r + 2$ more monopolized elements for this clique only when either this clique is a isolated clique or $M^{(n)}$ is itself a clique. For the latter case, in the statement of theorem 3.1 we use n elements to represent $G^{(n)}$ and note that $n^2/6 \geq n$ for $n \geq 6$. As for the former case, we lose all the edges joining this isolated clique to all the vertices not on this isolated clique, therefore we lose at least 5 edges from the calculated $\binom{n}{2}$ edges and hence further lose at least two cliques from the calculated $n^2/6$ cliques (as K_3). Besides, by lemma 3.2, we need use $r + 1$ more monopolized elements for this clique only when this clique has exactly $r + 2$ vertices of degree $(3 + r) - 1$. In this case, this clique has a vertex v adjacent to one vertex, say v' , not in this clique, and all vertices in this cliques other than v are not adjacent to v' . Therefore in $G^{(n)}$ we have $r + 2 \geq 3$ less edges than complete graph K_n , and thus we have still one less triangle in $\mathbf{Q}$. Now we have brought to the conclusion that in case that each Q_j has at least three edges, we never use more than $n^2/6$ elements in the statement of theorem 3.1 in order to form a representation of $G^{(n)}$.

Now we have secured a justification for assuming that some Q_j is an edge xy .

In case that $d(x) = d(y) = 1$, we can first form a representation of $G^{(n)} - x - y$ by the statement of theorem 3.1 using at most $\lfloor (n - 2)^2/4 \rfloor$ elements, and then use two new elements for the isolated edge xy to form a representation for $G^{(n)}$ using at most $\lfloor n^2/4 \rfloor$ elements. Thus in this case every representation of $G^{(n)}$ derived from $\mathcal{F}(\mathbf{Q})$, where $\mathbf{Q}$ is any greedy clique partition of $G^{(n)}$, by the statement of theorem 3.1 uses at most $\lfloor n^2/4 \rfloor$ elements.

As for the case that one of x, y has degree more than one, in any representation of $G^{(n)}$ derived from the statement of theorem 3.1 we can't use monopolized element on x or y . Now let R consist of the members of $\mathbf{Q} - \{Q_j\}$ that are incident to x , and S consist of those incident to y . Then the set

$$\mathbf{Q}' = \mathbf{Q} - (R \cup S \cup \{Q_j\})$$

is a greedy clique partition of

$$G' \equiv (G^{(n)} - x - y) - \bigcup_{Q_i \in R \text{ or } S} E(Q_i),$$

except possibly leaving some isolated vertices in M' uncovered by any members of $\mathbf{Q}'$. Recall that under the present case, in $\mathcal{F}$ we never use monopolize element on x, y . Now only if we can prove that every monopolize element in $\mathbf{S}(\mathcal{F})$ is always

necessary for deriving a representation of M' from $\mathcal{F}(\mathbf{Q}')$ by the statement of theorem 3.1, then by induction hypothesis we can prove that

$$|\mathcal{Q}(\mathcal{F}) - (R \cup S \cup \{Q_j\})| \leq \lfloor (n - 2)^2/4 \rfloor. \quad (1)$$

If in $\mathcal{F}$ we had used one monopolized element on some vertex v not belonging to any member of $R \cup S$, then in $\mathcal{F}(\mathbf{Q})$ $S_{\mathbf{Q}}(v)$ must be identical with some $S_{\mathbf{Q}}(u)$ where u is also a vertex not belonging to any member of $R \cup S$. Since both u and v don't belong to any member of $R \cup S$, in $\mathcal{F}(\mathbf{Q}')$ $S_{\mathbf{Q}'}(u) = S_{\mathbf{Q}'}(v)$. Thus this monopolized element is necessary for deriving a representation of M' from $\mathcal{F}(\mathbf{Q}')$ by the statement of theorem 3.1.

If in $\mathcal{F}$ we had used one monopolized element on some vertex v belonging to one member, say Q_v , of $R \cup S$, then in $\mathcal{F}(\mathbf{Q})$ $S_{\mathbf{Q}}(v)$ must be identical with some $S_{\mathbf{Q}}(u)$ where u is also a vertex belonging to Q_v . Now by lemma 3.2 v must have all its neighbors in Q_v . Thus v is an isolated vertex in G' . Thus this monopolized element is necessary for deriving a representation of G' from $\mathcal{F}(\mathbf{Q}')$ by the statement of theorem 3.1.

Now it suffices to prove that

$$|R \cup S| \leq n - 2,$$

since

$$n - 2 \leq \lfloor n^2/4 \rfloor - \lfloor (n - 2)^2/4 \rfloor - 1.$$

We prove this by choosing distinct vertices in $V(G) - \{x, y\}$ from the vertex sets of the members of $R \cup S$. Note that since each edge is covered exactly once in a clique partition, each $v \notin \{x, y\}$ appears once in R if v is adjacent to x and once in S if v is adjacent to y . Consider $Q_1 \in R$. If Q_1 contains a vertex v not adjacent to y , then we choose such a v for Q_1 . If all vertices in Q_1 are adjacent to y , then we choose for Q_1 a vertex $v \in Q_1$ such that vy belongs to the first member of $\mathbf{Q}$, say Q_2 , that contains both y and some vertex of Q_1 . Note that Q_2 is the only member of S containing v .

Now we have two cases, that is, either that Q_1 precedes xy in $\mathbf{Q}$ or that xy precedes Q_1 in $\mathbf{Q}$. For the first case, since Q_1 and xy are maximal when chosen, Q_2 must precedes Q_1 in $\mathbf{Q}$ for otherwise from the aforementioned hypothesis that all vertices in Q_1 are adjacent to y and Q_1 precedes xy in $\mathbf{Q}$, Q_1 should have contained y and hence xy . For the second case, since xy is maximal when chosen, one of Q_1, Q_2 precedes xy or otherwise xy should have contained v . Thus in this case Q_2 precedes

Q_1 in $\mathbf{Q}$. Note that in both above cases, we have that Q_2 precedes both of Q_1 , xy in $\mathbf{Q}$.

For the members of S , similarly as above choose vertices by reversing the roles of x and y .

In the above we have shown that if v belongs to some $Q_1 \in R$ and to some $Q_2 \in S$, and v is chosen for one of them, then the one for which it is chosen occurs after the other one in the ordered set $\mathbf{Q}$. Hence no vertex is chosen twice. Thus we conclude that

$$|R \cup S| \leq n - 2.$$

Therefore we are done. $\square$

Hence we have the following immediate consequence:

COROLLARY 3.3. *For a graph G with n vertices, $\omega(G) \leq \lfloor n^2/4 \rfloor$.*

4 Conclusion and Future Work

The edge clique partitions, as a special case of edge clique covers, are served as great classifying and clustering tools in many practical applications, therefore it is interesting to explore the concept in more details.

We keep working on the cases except multifamily and family, say antichain, uniform family etc. The results including greedy decompositions and also uniqueness of representations will be reported elsewhere. The greedy way to obtain these variants also gives the optimal upper bounds for the corresponding intersection numbers.

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